to particular environment. In other words, it is a purposeful manipulation of plant species in order to create desired genotype and phenotype for the benefit of humans. In early days, plant breeding activities were based mainly on skills and ability of person involved. But as the principles of genetics and cytogenetics have elucidated breeding methods such as selection, introduction, hybridization, ploidy, mutation, tissue culture and biotechnology techniques were designed to develop improved crop varieties.

9.6.1. Objectives of Plant Breeding

- To increase yield, vigour and fertility of the crop
- To increase tolerance to environmental condition, salinity, temperature and drought.
- To prevent the premature falling of buds, fruits etc.
- To improve synchronous maturity.
- To develop resistance to pathogens and pests.
- To develop photosensitive and thermos-sensitive varieties.

9.6.2. Steps in Plant Breeding

The main steps in plant breeding are given below

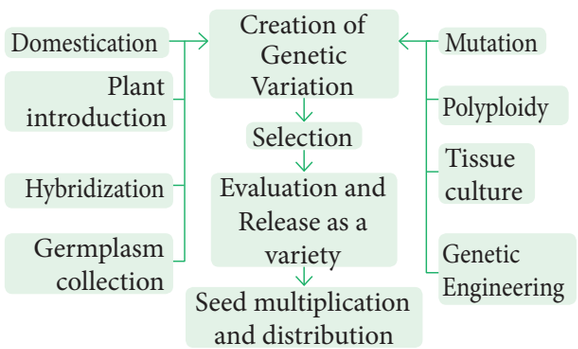

Figure 9.11 : Steps in Plant Breeding

9.7 Conventional Plant Breeding Methods

Conventional plant breeding methods resulting in hybrid varieties had a tremendous impact on agricultural productivity over the last decades. It develops new plant varieties by the process of selection and seeks to achieve expression of genetic material which is already present within the species. In this chapter we will discuss about some of the conventional methods of plant breeding.

9.7.1. Plant Introduction

Plant introduction may be defined as the introduction of genotypes from a place where it is normally grown to a new place or environment. Rice variety of IR8 introduced from Philippines and Wheat varieties of Sonora 63, Sonora 64 from Mexico.

The newly introduced plant has to adapt itself to the new environment. This adjustment or adaptation of the introduced plant in the changed environment is called **acclimatization.** All the introductions must be free from presence of weeds, insects and disease causing organisms. This has to be carefully examined by the process called **quarantine**, a strict isolation imposed to prevent the spread of disease.

Introduction may be classified as Primary introduction and Secondary introduction

(1) **Primary introduction** - When the introduced variety is well adapted to the new environment without any alternation to the original genotype.

(2) **Secondary introduction** - When the introduced variety is subjected to selection to isolate a superior variety and hybridized with a local variety to transfer one or a few characters to them. The botanical gardens in different parts of the world also played a significant role in plant introduction. Example : Tea varieties collected from China and North East India initially grown in Botanical Garden of Kolkata from which appropriate clones have selected and introduced to different parts of India.

9.7.2. Selection

Selection is the choice of certain individuals from a mixed population for a one or more desirable traits. Selection is the oldest and basic method of plant breeding. There are two main types of Selection.

i. **Natural Selection:** This is a rule in nature and results in evolution reflected in the Darwinian principle "survival of the fittest". It takes longer time in bringing about desired variation.

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